



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 64: TI Solar Battery Pack

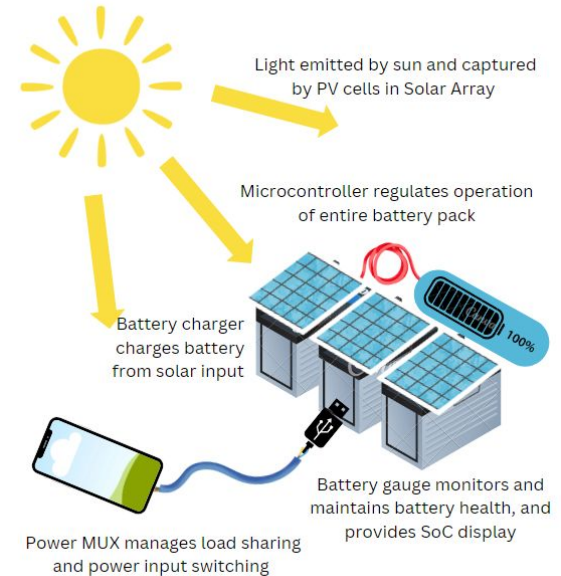
Jaein Cha, Patrick Wang, Cole Miller, Camden Pomeroy

Sponsor: Wyatt Keller

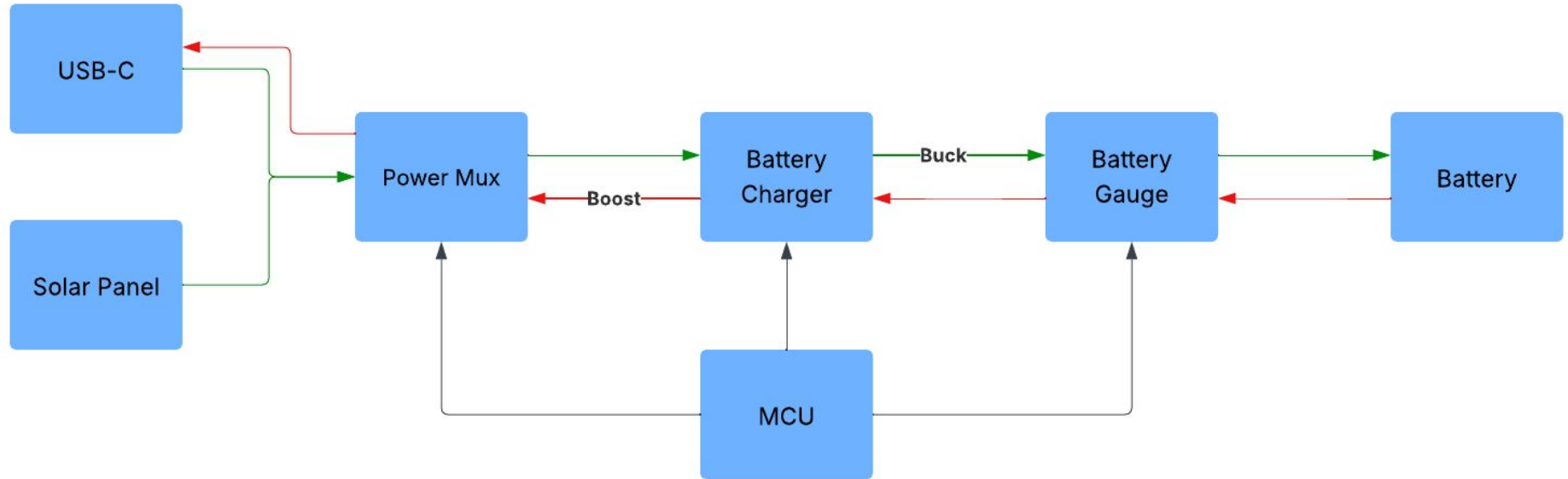
TA: Ali Alenezi

Project Summary

- Natural disasters, camping, or remote backpacking may result in a scenario with the inability to charge cell phones and other necessary small electronic devices.
- A solution is to create a solar powered battery pack that charges itself from solar panels, and distributes the power to electronic devices through a USB connection.



Subsystem Overview

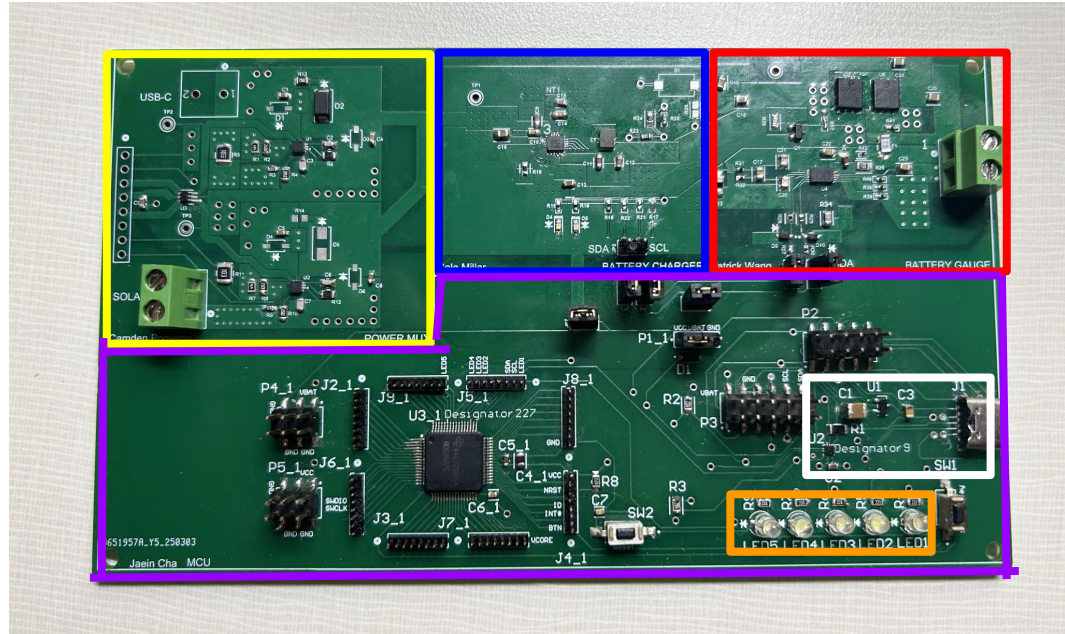


Integrated System Design

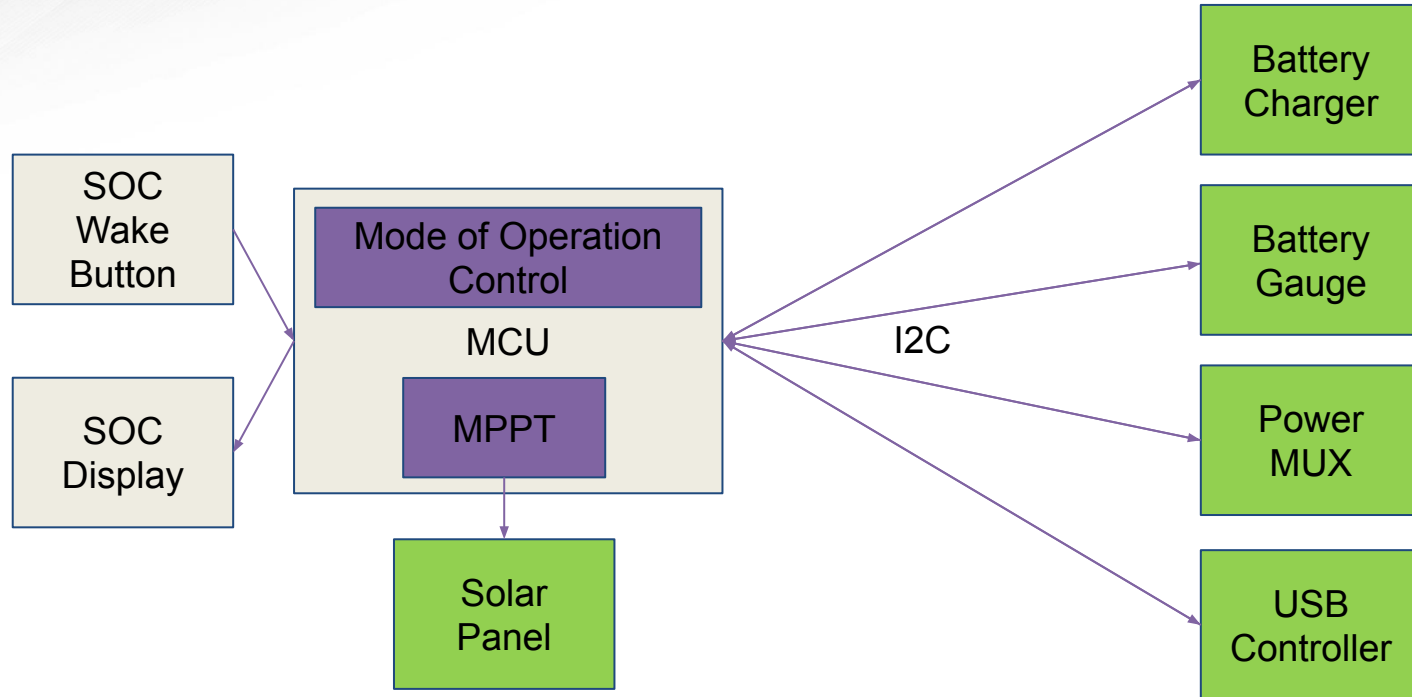


Board Design

Yellow - Power Mux
Blue - Battery Charger
Red - Battery Gauge
Purple - MCU
Orange - SOC Display
White - USBC system

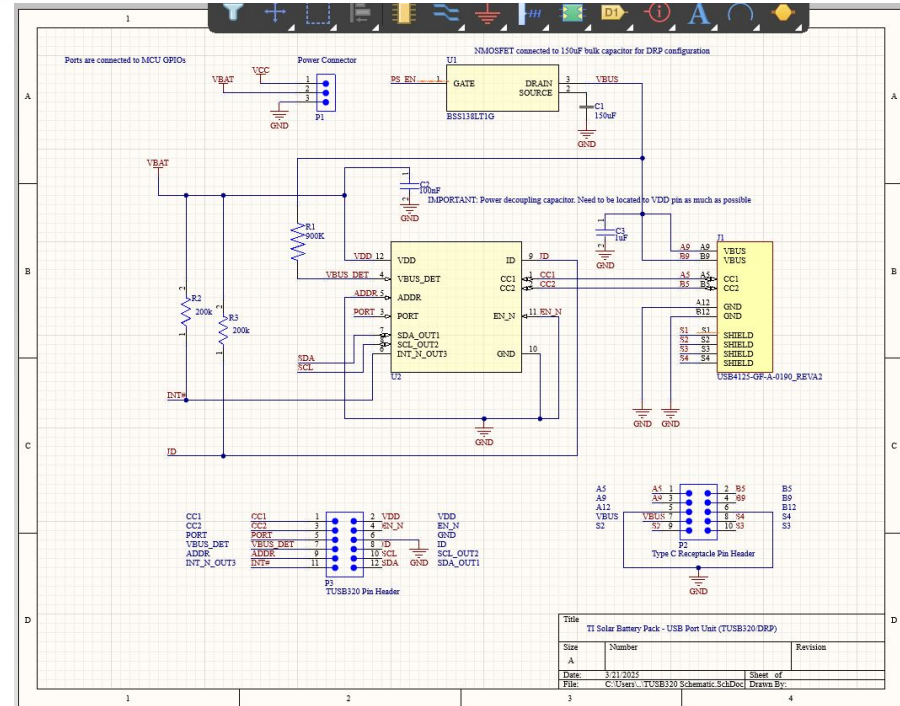


Integrated System Diagram (Software)



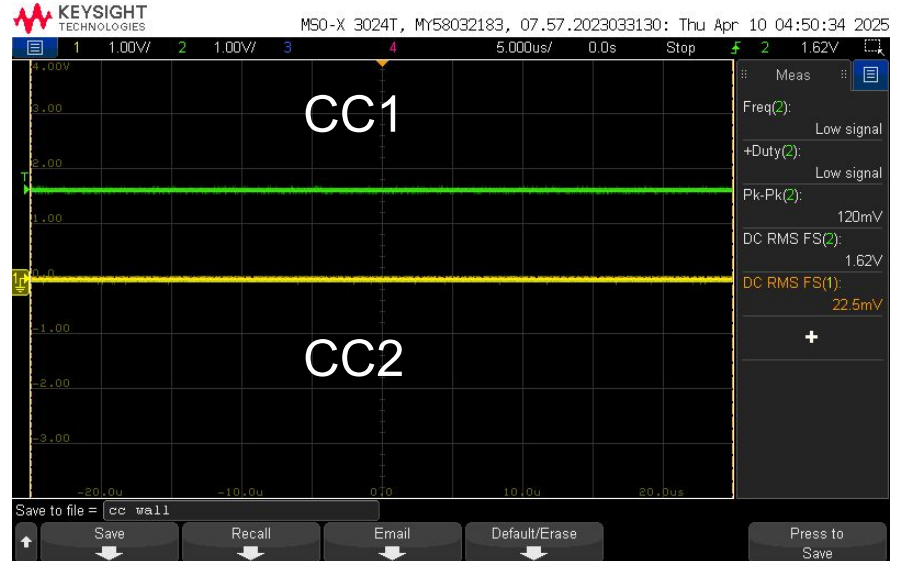
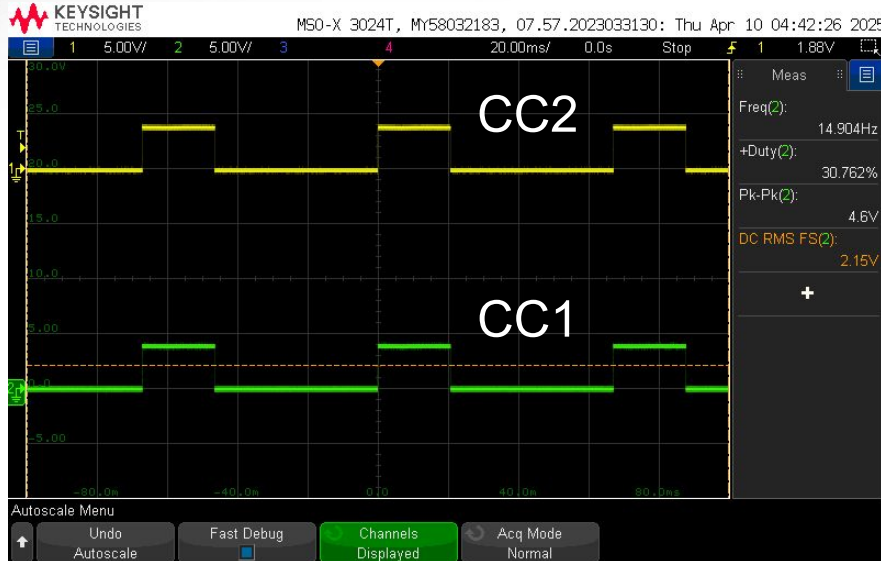
USB-C Input

- USB-C logic controlled by TUSB320
- Configured as a DRP in I2C mode
- Validated to source and sink devices and control VBUS



USB-C Input

- CC lines when connected to a sink (Left), and when connected to a source (Right)



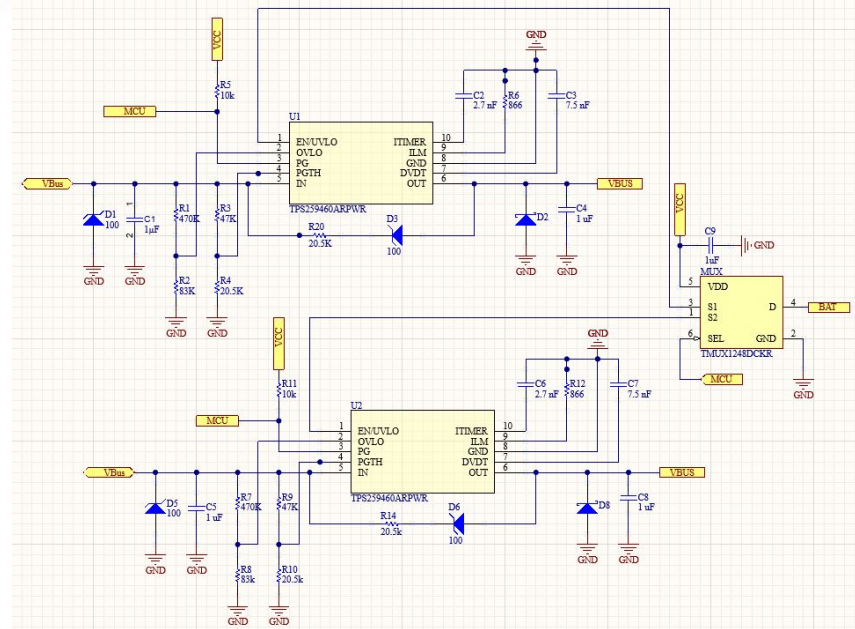
Solar Panel Input

- 2.5V to 7.5V output at 15W across three foldable lightweight panels
- On a standard sunny day charges at 7V 2A on average
- Cloudy day 7V at 0.5A

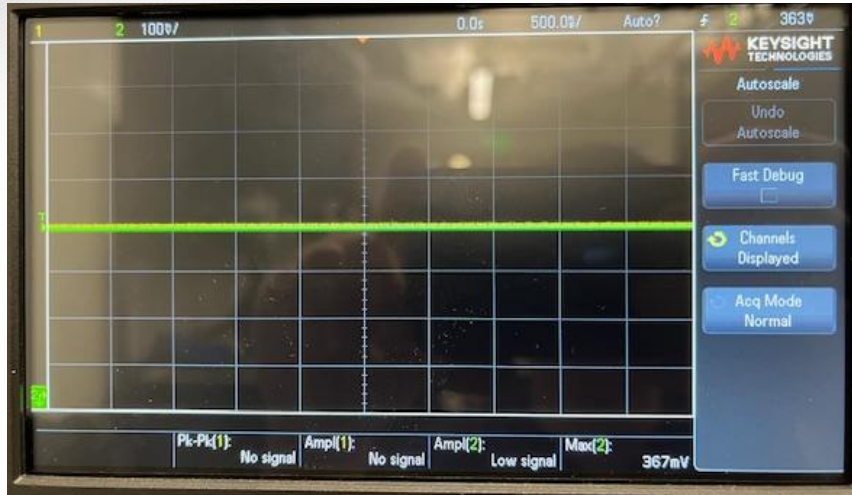


Power Mux

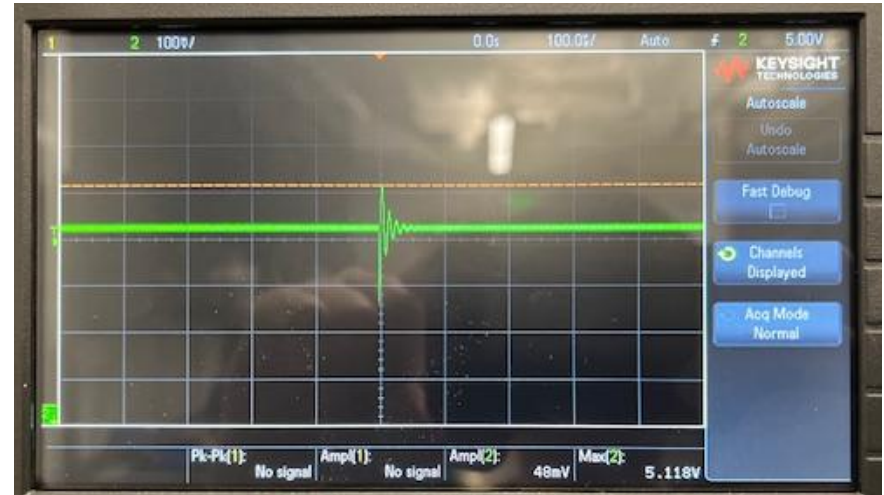
- Protection of system when USB-C and Solar inputs are connected
- Break before make switching
- UV/OV Shutoff Protection with Auto Retry
- MCU good input power detection and control over enabling power paths



Power Mux



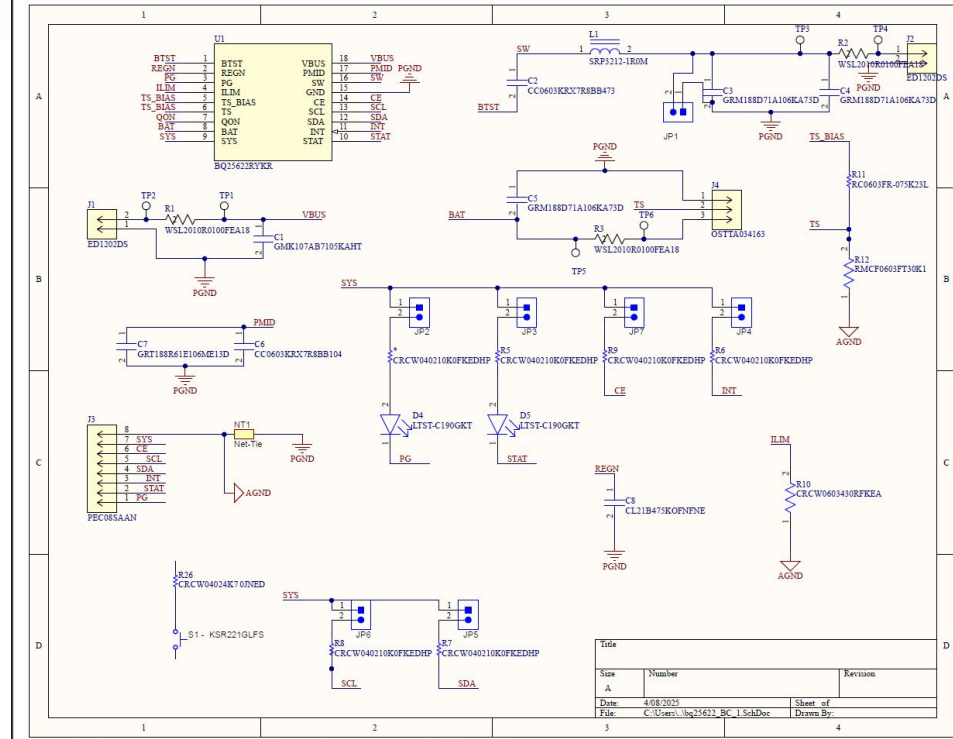
5V on USB-C and EN low on eFuse
~0V output



5V on USB-C and EN high on eFuse
~5V output

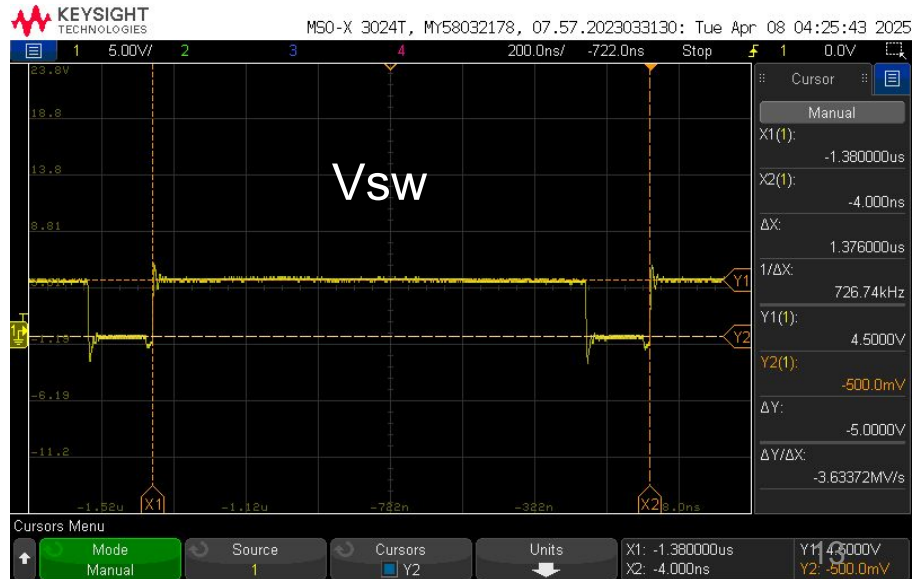
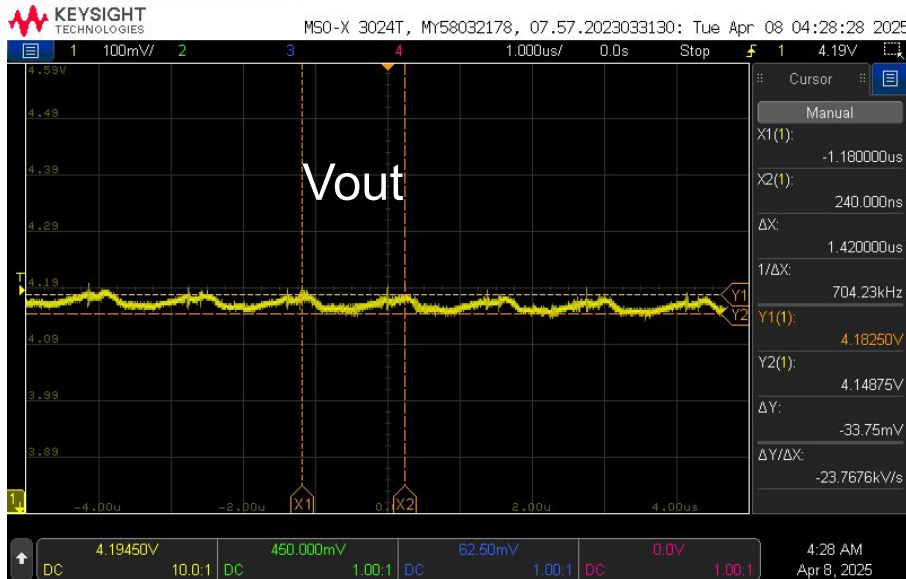
Battery Charger

- Utilizes a BQ25622
- While charging battery acts as a buck converter
- While charging Devices acts as a boost converter
- Controls current and voltage



Battery Charger - Buck

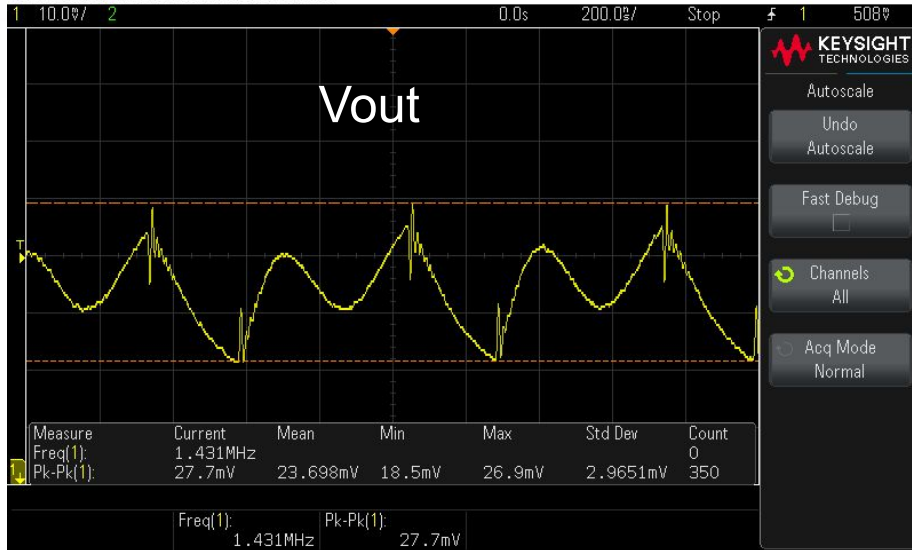
Depicted on the left is V_{out} at 4.2V with a 40 mV ripple. On the right is the switching voltage.



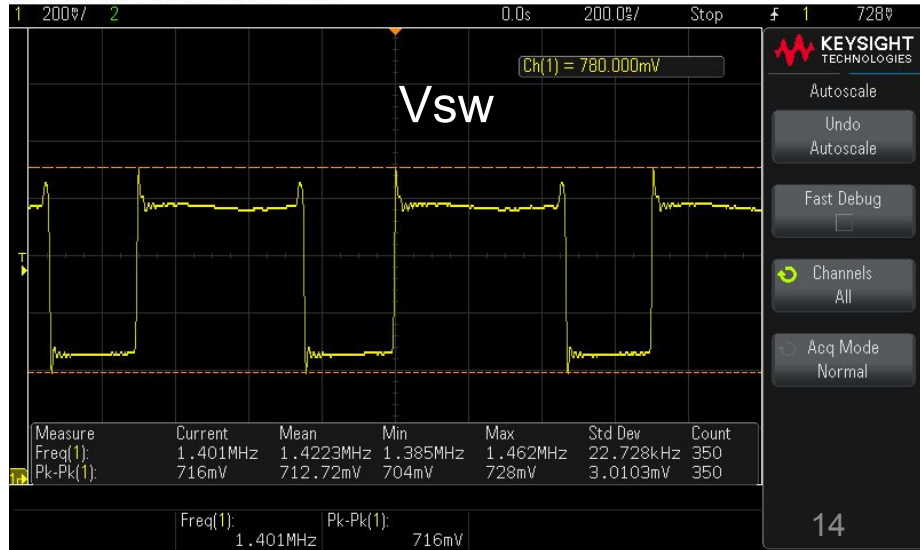
Battery Charger - Boost

Depicted on the left is the AC portion of V_{out} at 5V with a 20 mV ripple. Depicted on the right is the switching voltage.

EDUX10526, CN63131164: Sat Mar 29 04:52:21 2025

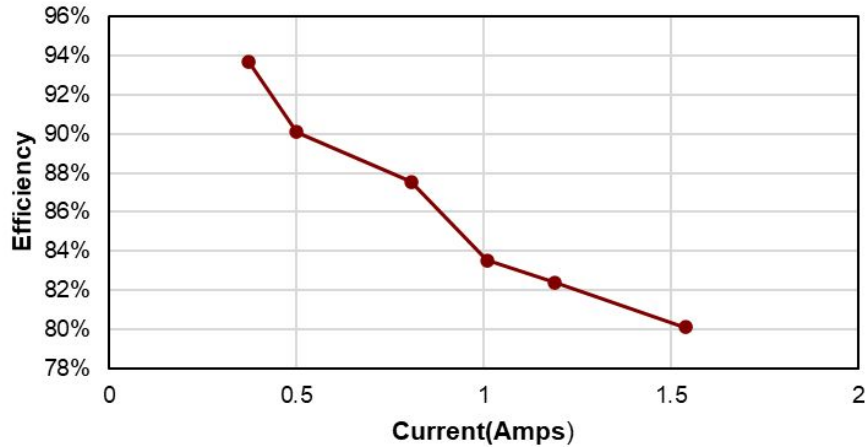


EDUX10526, CN63131164: Sat Mar 29 04:51:47 2025

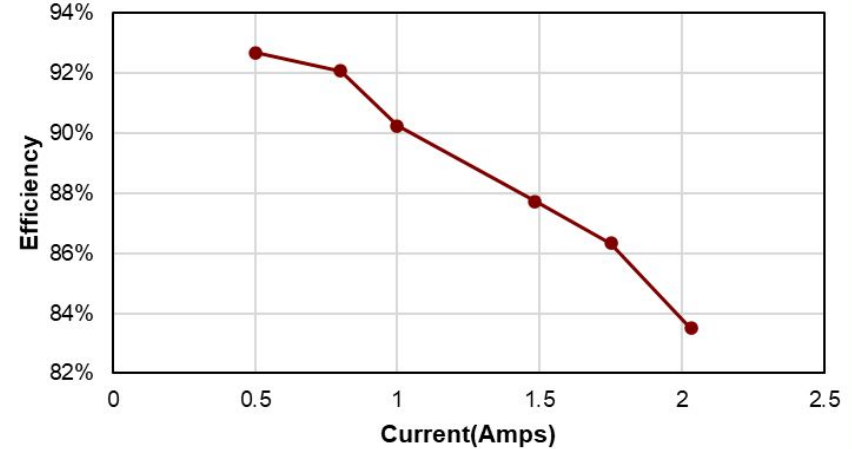


Battery Charger - Efficiency

Boost Efficiency

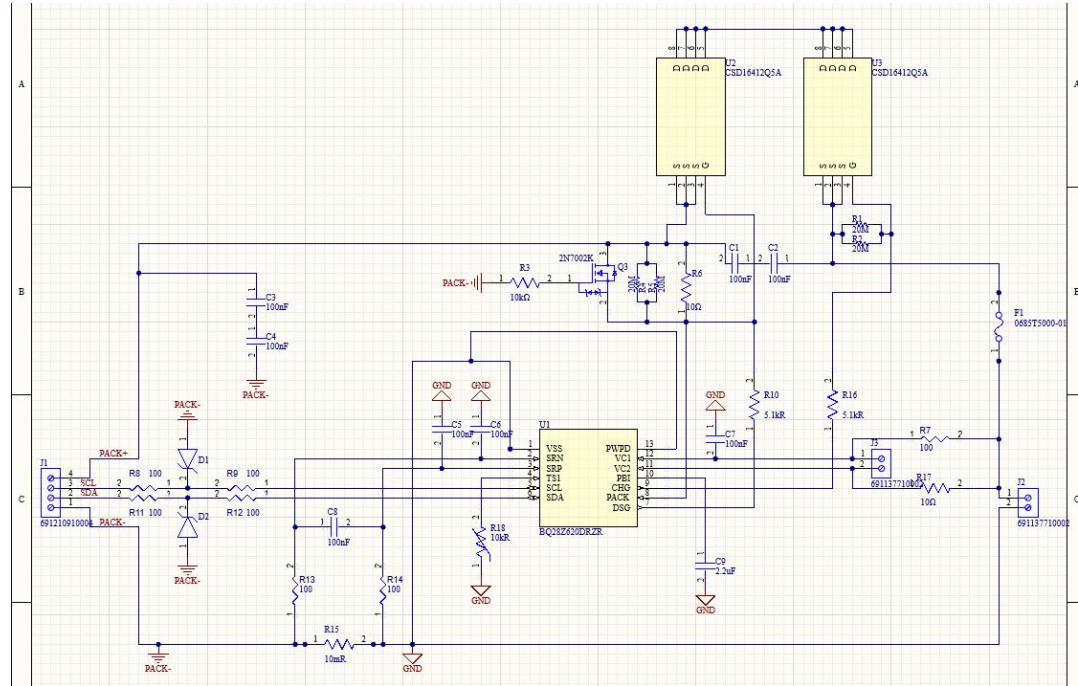


Buck Efficiency



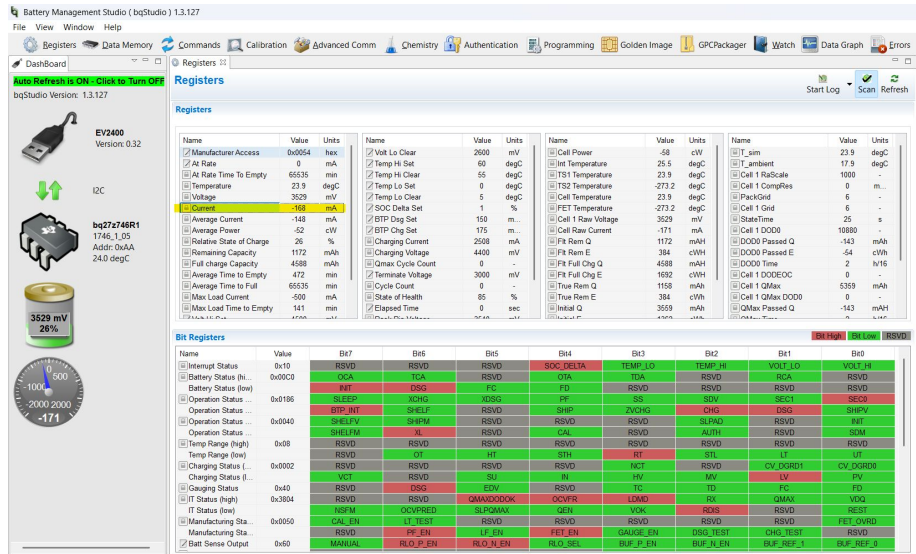
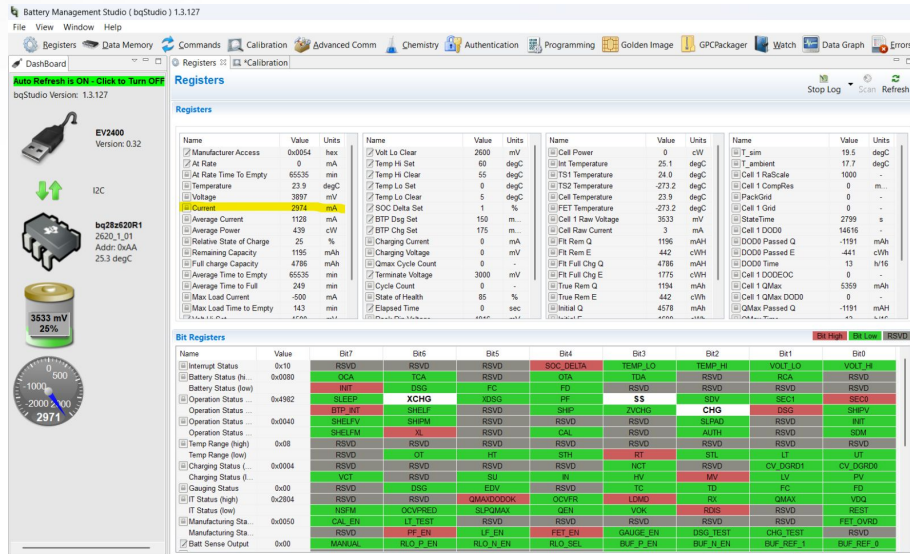
Battery Gauge

- Utilizes bq28z620
- Monitors voltage, current, temperature, and calculates state-of-charge (SOC) for Li-ion pack
- Configurable overvoltage (OV) and undervoltage (UV) thresholds to protect against faults
- Triggers fault flags (e.g., OV/UV, overcurrent) in registers



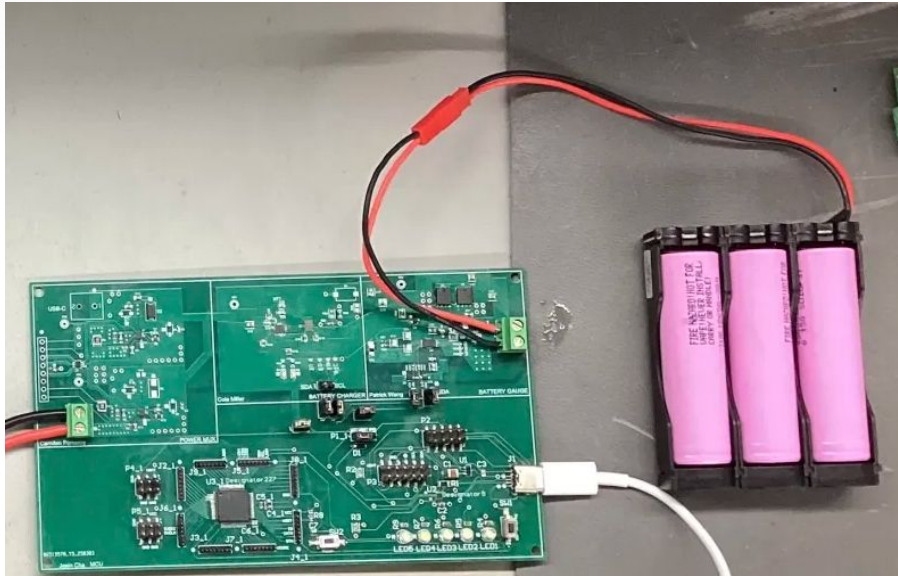
Input 4V 3A

Input 2V 3A



Battery Pack

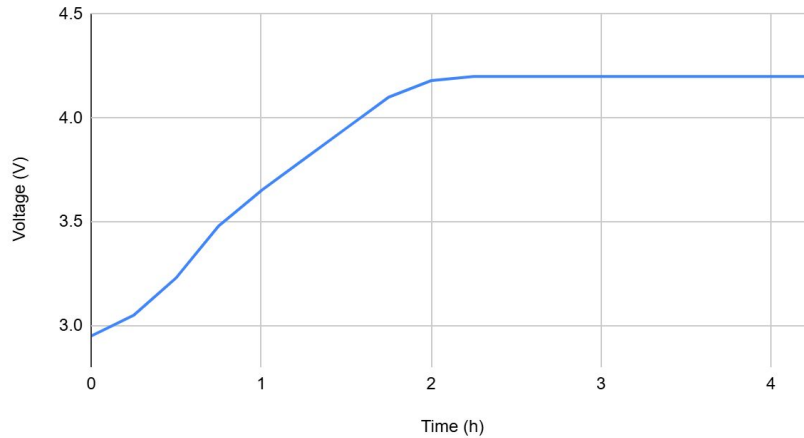
1S3P configuration using Samsung INR18650 30Q



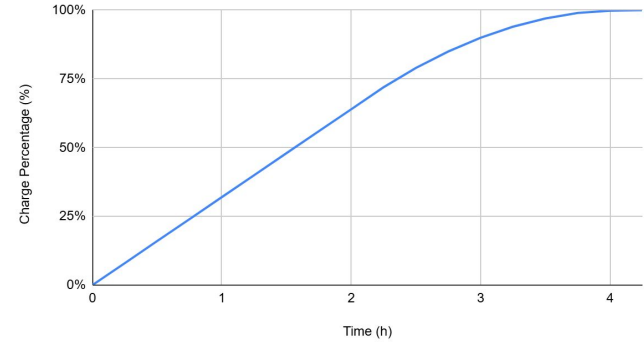
Type = Li-ion
Nominal voltage = 3.7V
Design capacity (1) = 3000mah
Charge Voltage: 4.2V
Discharge Cut-off Voltage: 2.5V

Battery Pack (charging)

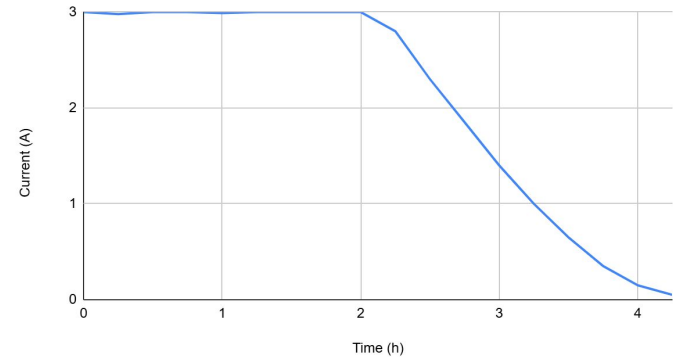
Voltage (V) vs. Time (h)



State of Charge (%) vs. Time (h)

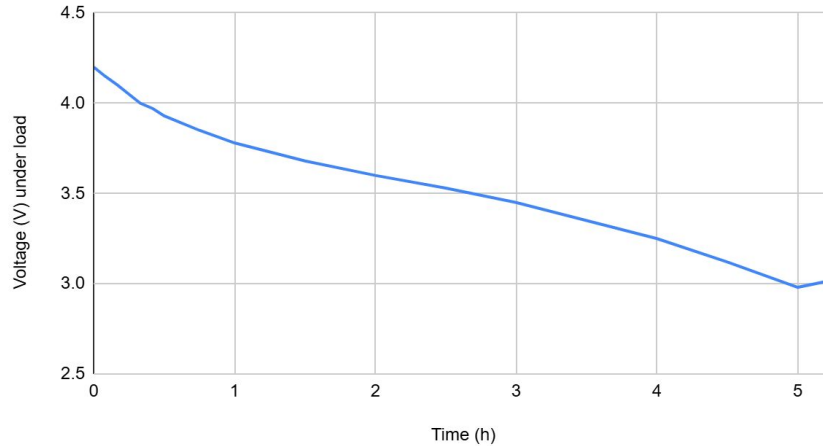


Current (A) vs. Time (h)

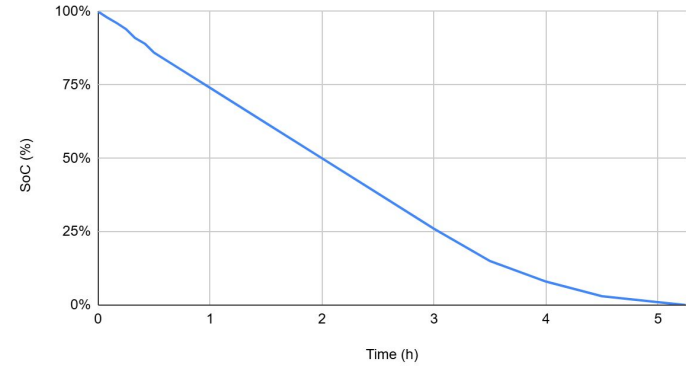


Battery Pack (discharging)

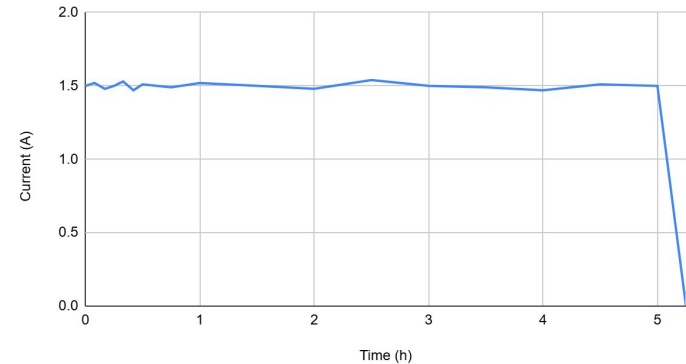
Voltage (V) under load vs. Time (h)



SoC (%) vs. Time (h)



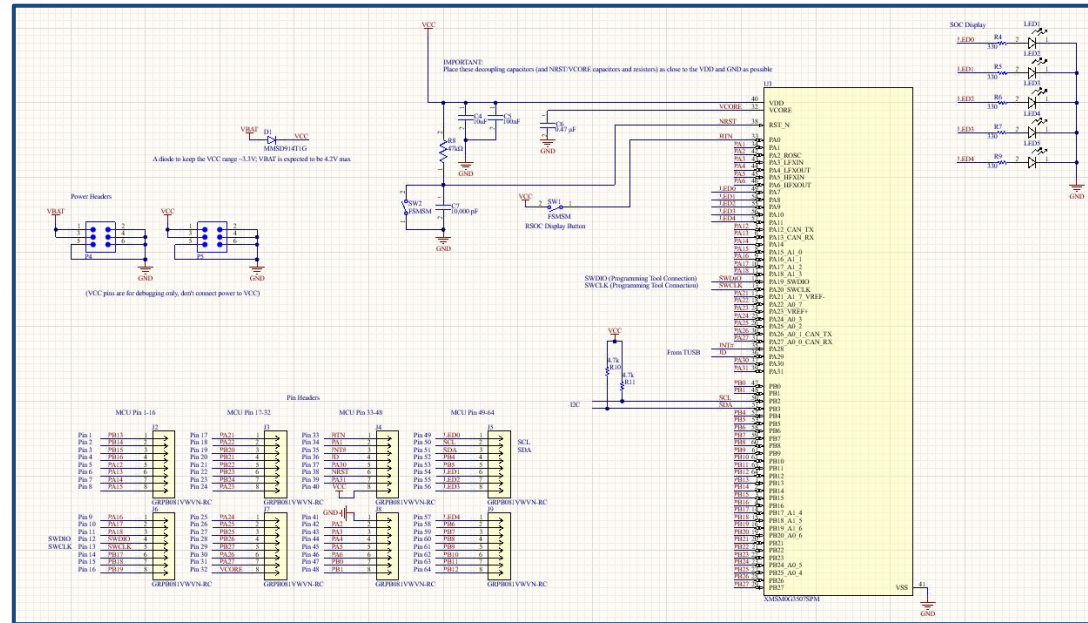
Current (A) vs. Time (h)



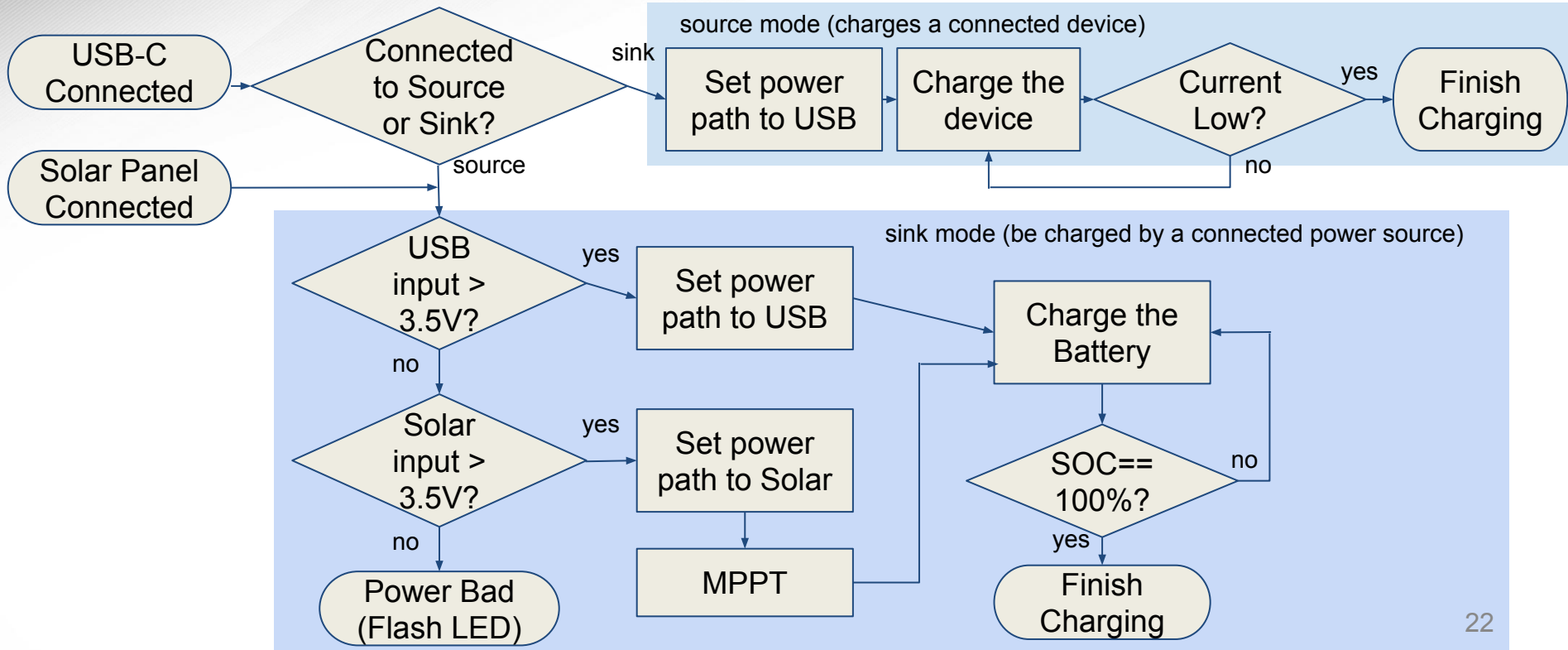
MCU

- Control unit - integrates subsystems and firmware into one organized system
- MSPM0G3507 (LQFP 64 pin)
- I2C Communication *Inter-Integrated Circuit*
 - Read/write register values for subsystem control
- MPPT *Maximum Power Point Tracking*
 - To maximize power from solar panel
- SOC display and SOC wake button

SOC: *State of Charge*

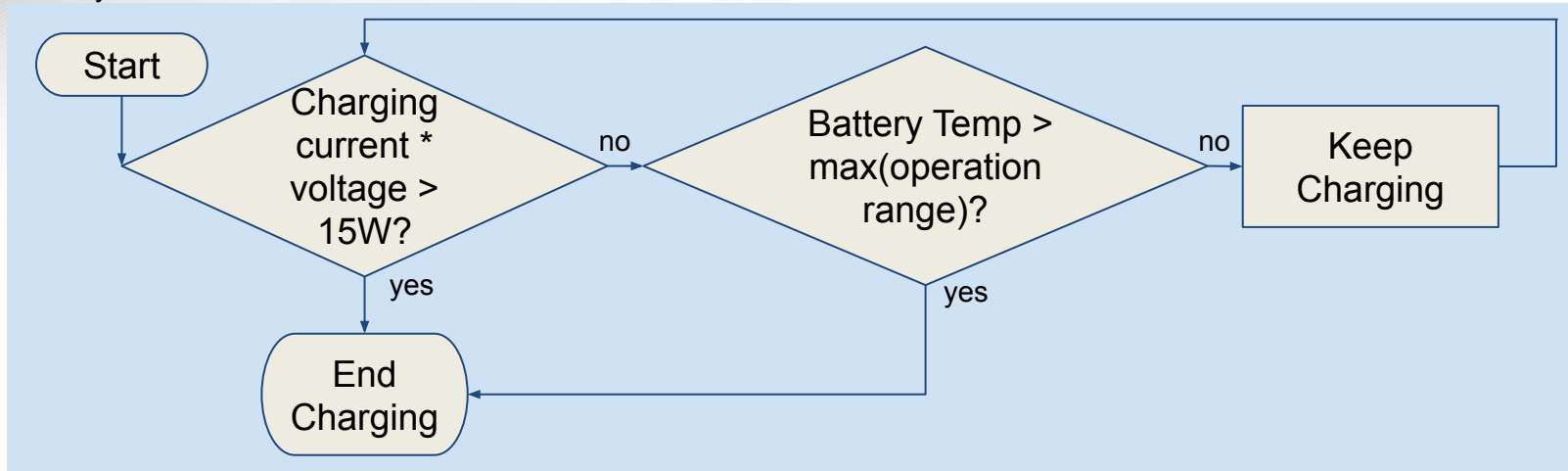


MCU Firmware Flow Chart

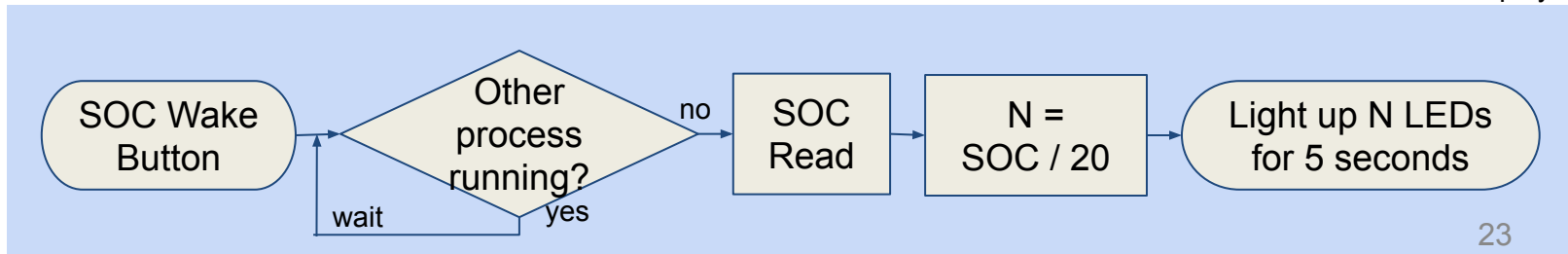


MCU Firmware Flow Chart

safety ensurement



SOC Display



System Summary

- Utilizing a standard USB-C input of 5V 3A charging the batteries at 3.7 V will take 2 hours and 45 minutes.
- Utilizing the solar panel on a sunny day generating 7V 2A, charging the batteries at 3.7V will take approximately 3 hours.
- Utilizing the solar panel on a cloudy day generating 7V 0.5A, charging the batteries at 3.7V will take approximately 10.5 hours.

- Discharging the battery at 2A will take approximately 4 hours and 30 minutes
- Discharging the battery at 1.5A will take approximately 5 hours and 15 minutes

- Has capacity to fully charge 2.7 iphone 14s
- Devices consume 1.2795 mA during operation
- Devices consume 547.1 μ A while asleep (monitoring battery)
- Devices consume 2.1 μ A while off

SYSTEM Validation Plan

Number	Validation Test	Input	Expected Output	Results	Status	Engineer	Reference Number
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	As expected		Camden Pomeroy	3.2.3.2.2
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected		Camden Pomeroy	3.2.3.2.2
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested		Camden Pomeroy	6.1.1
4	PP1 PG (Bad signal)	If Vin < 2V	PG signal is read High from MCU	Not tested		Camden Pomeroy	6.1.1
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested		Camden Pomeroy	6.1.1
6	PP2 PG (Bad signal)	If Vin < 2V	PG signal is read High from MCU	Not tested		Camden Pomeroy	6.1.1
7	PP1 UVLO Operational	If Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
8	PP2 UVLO Operational	If Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
9	PP1 OVLO Operational	If Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
10	PP2 OVLO Operational	If Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
11	SEL USBC	SEL is sent low from MCU	PP1 is active and PP2 is not	Not tested		Camden Pomeroy	6.1.4
12	SEL Solar	SEL is sent high from MCU	PP2 is active and PP1 is not	Not tested		Camden Pomeroy	6.1.4
13	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected		Camden Pomeroy	3.2.3.2.2
14	SOC Display	SOC from 0-100%	1,2,3,4, or 5 LEDs on as expected	Not tested		Jaein Cha	3.2.3.2.1
15	I2C functionality (1:1, EVM)	Run I2C test code	It reads the default register value	As expected		Jaein Cha	6.1
16	I2C functionality (1:1, on-board)	Run I2C test code	It reads the default register value	Not tested		Jaein Cha	6.1
17	I2C functionality (1:N, EVM)	Run I2C test code	It reads the default register values	As expected		Jaein Cha	6.1
18	I2C functionality (1:N, on-board)	Run I2C test code	It reads the default register values	Not tested		Jaein Cha	6.1
19	I2C - initialization	3.3V Power supply to the pc	Setup the communication correctly	As expected		Jaein Cha	6.1
20	I2C - DFP configuration	DFP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
21	I2C - UFP configuration	UFP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
22	I2C - DRP configuration	DRP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
23	Reset	Reset button pressed	MCU resets and stop working whi	Not tested		Jaein Cha	3.2.3.2.1
24	MPPT (simulation)	Various voltages	maintain the power as close as 15	As expected		Jaein Cha	6.2
25	MPPT (on-board)	Various voltages	maintain the power as close as 15	Not tested		Jaein Cha	6.2
26	SOP Display	SOP button	Lights up all the LEDs when SOP	Not tested		Jaein Cha	3.2.3.2.1
27	Standard Operation	5V 3A on VBUS	around 5V 3A on VBUS	4.2V 3.2A on VBAT		Cole Miller	2.1
28	USB OTG Function	4.2V 3.2A on VBAT	around 4.2V 3.2A on VBAT	5V 1.5A on VBUS		Cole Miller	2.2
29	ILIM Register Functionality	If input current >1A	output current >1A	Output 1A		Cole Miller	2.3
30	VLM Functionality	If Vin > 4.2V	Vo > 4.2V	Output 4.2V		Cole Miller	2.4
31	Solar Panel VOC	Solar panel Voc < 18V	Solar panel Voc < 18V	As expected		Cole Miller	6.3.1
32	Battery Charger I2C	Write to registers. Write cha	Read from said registers	As expected		Cole Miller	6.1.2
33	Solar Panel Power	regular usage in sunny day	output 15W	As expected		Cole Miller	6.3.12
34	Battery Gauge I2C	Write to registers. Read batt	Read from said registers	As expected		Patrick Wang	6.1.3
35	Battery Lifespan	Battery connected to load	The lifespan is at least 3 hours	As expected		Patrick Wang	3.2.1.2
36	Power Consumption	Normal Operating Condition	Maximum power consumption is t	As expected		Patrick Wang	3.2.1.2
37	Battery SOH	Discharge batteries from 10	Batteries stay in within an approx	As expected		Patrick Wang	6.1
38	High Power Path Charge	5V 3A from power MUX	Charges Battery Pack	As expected		Patrick, Cole, Camden	3.2.3.2.2
39	High Power Path Discharge	Load Connected to USBC	Battery Pack Discharges 15W	Not tested		Patrick, Cole, Camden	3.2.3.2.2
40	USBC Charging	USBC plug from wall outlet	Battery Pack Charges	Not tested		Patrick Wang	6.2.1



Questions?